Ratio & Proportion

Year 4 · **Homework Worksheet**

Worked Example

Worked example: Share 20 marbles in the ratio 1:4.

Answer: 1+4=5 parts. 20÷5=4 per part. Shares: 4 and 16.

Easy / Standard (15 questions)

1. Scale up the ratio 3:2 by multiplying both parts by 2. What is the new ratio? A) 5:4 B) 6:4 C) 6:2 D) 3:4
2. Share 4 sweets equally between 2 people. Select ALL statements that are TRUE: A) Each person gets 2 sweets B) Each person gets 4 sweets C) This is sharing in the ratio 1:1 D) This is sharing in the ratio 2:1 E) Together they get 4 sweets
3. A recipe for 1 person needs 8 biscuits. How many biscuits for 4 people?
4. A pattern uses 3 blue beads for every 1 red bead. If there are 5 red beads, how many blue beads are there?
5. A recipe needs 3 eggs for every 1 cake. How many eggs are needed for 4 cakes?
6. Scale up the ratio 4:4 by multiplying both parts by 4. What is the new ratio? A) 8:8 B) 16:4 C) 16:16 D) 4:16
7. Share 6 sweets equally between 2 people. How many does each get?
8. A recipe for 1 person needs 6 biscuits. How many biscuits for 3 people?
9. A pattern uses 3 blue beads for every 1 red bead. If there are 3 red beads, how many blue beads are there?
10. A recipe needs 3 eggs for every 1 cake. How many eggs are needed for 6 cakes?
11. Scale up the ratio 4:2 by multiplying both parts by 3. What is the new ratio? A) 7:5 B) 12:6 C) 12:2 D) 8:6
12. Share 10 sweets equally between 2 people. Select ALL statements that are TRUE: A) Each person gets 5 sweets B) Each person gets 10 sweets C) This is sharing in the ratio 1:1 D) This is sharing in the ratio 2:1 E) Together they get 10 sweets
13. A recipe for 1 person needs 7 biscuits. How many biscuits for 2 people?
14. A pattern uses 2 blue beads for every 1 red bead. If there are 3 red beads, how many blue beads are there?
15. A recipe needs 1 egg for every 1 cake. How many eggs are needed for 5 cakes?

Standard (15 questions)

1. Share 16 counters in the ratio 3:1. Select ALL statements that are TRUE: A) The shares are 12 and 4 B) The shares are 8 and 8 C) There are 4 parts in total D) There are 3 parts in total E) The larger share is 12
2. In a fruit bowl, apples and oranges are in the ratio 1:1. There are 4 pieces of fruit in total. How many apples are there?
3. A model car is built at a scale where 1 cm represents 16 cm in real life. If the model is 5 cm long, how long is the real car? A) 21cm B) 60cm C) 80cm D) 90cm
4. A recipe for 3 people uses 9g of sugar. How much sugar is needed for 9 people?
5. A recipe for 1 person uses 3 grams of butter. How much butter is needed for 3 people?
6. Share 18 counters in the ratio 3:3.

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7. In a fruit bowl, apples and oranges are in the ratio 1:1. There are 6 pieces of fruit in total. How many apples are there?

8. A model car is built at a scale where 1 cm represents 17 cm in real life. If the model is 5 cm long, how long is the real car? A) 22cm B) 65cm C) 85cm D) 90cm

9. A recipe for 2 people uses 6g of sugar. How much sugar is needed for 6 people?

10. A recipe for 1 person uses 5 grams of butter. How much butter is needed for 4 people?

11. Share 16 counters in the ratio 1:3. Select ALL statements that are TRUE: A) The shares are 4 and 12 B) The shares are 8 and 8 C) There are 4 parts in total D) There are 3 parts in total E) The larger share is 12

12. In a fruit bowl, apples and oranges are in the ratio 3:1. There are 16 pieces of fruit in total. How many apples are there?

13. A model car is built at a scale where 1 cm represents 18 cm in real life. If the model is 2 cm long, how long is the real car? A) 20cm B) 36cm C) 40cm D) 18cm

14. A recipe for 4 people uses 12g of sugar. How much sugar is needed for 12 people?

15. A recipe for 1 person uses 5 grams of butter. How much butter is needed for 6 people?

Standard / Advanced (15 questions)

1. Boys and girls in a club are in the ratio 1:1, totalling 10 children. How many more of the larger group are there than the smaller group? A) 0 B) 2 C) 5 D) 10

2. If 1 box holds 4 cakes, how many cakes are in 5 boxes?

3. Simplify the ratio 6:9 to its simplest form. A) 3:2 B) 2:3 C) 6:9 D) 2:9

4. Share 20 in the ratio 3:1. Find the difference between the two shares.

5. A cake recipe for 4 people needs 6g of flour. Chloe wants to make the cake for 8 people. How much flour is needed?

6. A hose fills a paddling pool at 5 litres per minute. After 2 minutes it has filled 10 litres. The pool needs 30 litres to be full. Select ALL statements that are TRUE: A) The pool needs 6 minutes in total to fill B) 4 more minutes are needed from this point C) 20 more litres are needed D) 5 more minutes are needed from this point E) The rate is 5 litres per minute

7. A paint mix uses red and yellow in the ratio 1:1, using 10 litres in total for one wall. How many litres of red paint are needed?

8. On a model railway, 1 cm represents 20 cm in real life. A model bridge is 4 cm long. Select ALL statements that are TRUE: A) The scale is 1cm to 20cm B) The real bridge is 80cm long C) The real bridge is 60cm long D) The scale is 1cm to 4cm E) The real bridge is 100cm long

9. A fruit basket has apples and pears in the ratio 1:1, with 10 pieces of fruit altogether. If each apple costs £1, how much would all the apples cost?

10. A cake recipe for 3 people needs 6g of flour. Ryan wants to make the cake for 6 people. How much flour is needed?

11. A hose fills a paddling pool at 5 litres per minute. After 4 minutes it has filled 20 litres. The pool needs 35 litres to be full. How many MORE minutes will it take from this point?

12. A paint mix uses red and yellow in the ratio 1:2, using 9 litres in total for one wall. How many litres of red paint are needed? A) 2 litres B) 3 litres C) 4 litres D) 6 litres

13. On a model railway, 2 cm represents 30 cm in real life. A model bridge is 4 cm long. How long is the real bridge?

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14. A fruit basket has apples and pears in the ratio 1:2, with 9 pieces of fruit altogether. If each apple costs £2, how much would all the apples cost?

15. A recipe for 2 people uses 6g of rice. How much for 6 people (triple the original)?

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